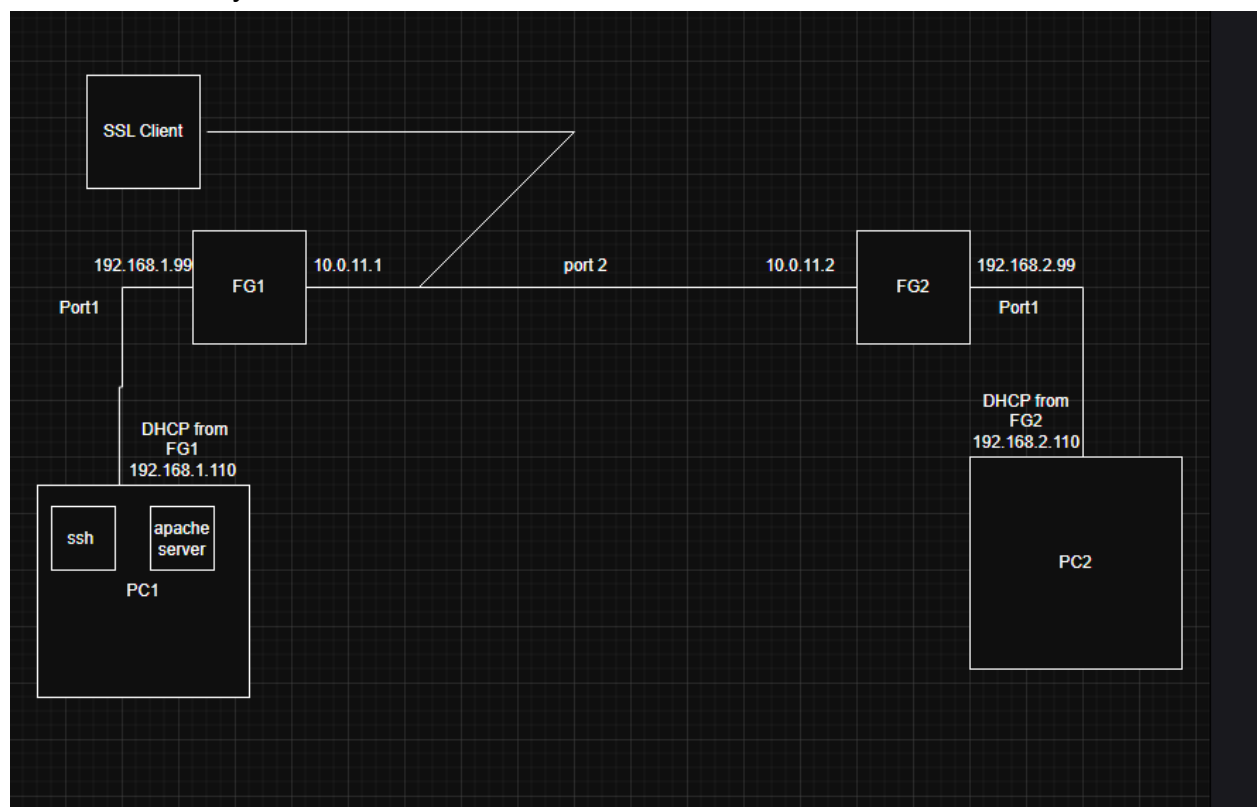


The network design features two core geographic locations interconnected across a transit interface, along with provisioning for a secure remote workforce. The primary internal headquarters LAN is built out behind a Fortinet FortiGate appliance, designated as FG1, which serves as the default gateway for PC1. Within this primary subnet, PC1 is connected to FG1 with an IP address of 192.168.1.99 for out-of-band management and direct administrative access to the firewall's graphical user interface. To facilitate dynamic host configurations for standard network endpoints, a Local Area Network (LAN) Dynamic Host Configuration Protocol (DHCP) server was configured on FG1 to distribute addresses within the 192.168.1.1 to 192.168.1.255 range. To maintain administrative access and prevent address collisions with static assets, an exclusion pool was strictly defined from 192.168.1.90 through 192.168.1.99.

the secondary branch location operates under a mirror topology managed by a second Fortinet appliance, designated as FG2. For this site, PC2 was configured with an IP address of 192.168.2.99. The Port 1 address on FG2 was modified to act as the internal gateway interface, changing its address space to correctly bind to the secondary LAN network. The local DHCP scope on the FG2 port was subsequently adjusted to hand out the appropriate matching IP network range, enabling newly connected branch hosts to successfully acquire dynamic IP leases, resolve their default gateways, and achieve local connectivity.



Implement Site-to-Site VPN

A Site-to-Site IPsec Virtual Private Network tunnel was established between the appliances. The interconnect layout leverages Port 2 on both firewalls as the dedicated outbound public-facing WAN interface for the remote IP outgoing interface connection. During the network design phase, FG1 was assigned a local IP of 10.10.11.1 and FG2 was assigned a local IP of 10.10.11.2. For security policy definition, the remote subnet configuration for FG1 was mapped to the branch network 192.168.2.0, while the remote subnet configuration for FG2 was mapped back to the primary HQ network 192.168.1.0 using the Internet Key Exchange version 1 (IKEv1). To bring the tunnel online, active ICMP echo requests IPsec Monitor VPN tunnel dashboard, the connection was brought up manually by activating it, forcing an active status change and successfully permitting encrypted bidirectional traffic flow between the two networks.

The screenshot shows the Fortinet Firewall Policy configuration page. The left sidebar contains a navigation menu with options like Dashboard, Network, Policy & Objects, and System. The 'Policy & Objects' section is expanded, showing a list of firewall policies. The main area displays a table of firewall policies with columns: Name, Source, Destination, Schedule, Service, Action, NAT, Security Profiles, Log, and Bytes. The 'Implicit' policy is highlighted in blue. The table lists various policies such as 'FG1 to FG2', 'VPN-FG1 to FG2_remote_0', 'DOS policy', 'Internal to FG1 to FG2', 'VM to Host', 'Internet to WAN1', 'SSL-VPN tunnel interface (sslroot)', 'SSL to DMZ', 'wan1', and 'Implicit'. Each policy row shows details like Name, Source, Destination, Schedule, Service, Action, NAT, Security Profiles, Log, and Bytes.

The top screenshot shows the FortiGate web interface with the 'VPN' menu expanded. The 'IPsec Tunnels' option is selected. The main content area displays a table of IPsec tunnels:

Tunnel	Interface Binding	Status	Ref.
FG1 to FG2	Lan2	Up	6

The bottom screenshot shows the 'Edit VPN Tunnel' configuration page for the 'FG1 to FG2' tunnel. The configuration is as follows:

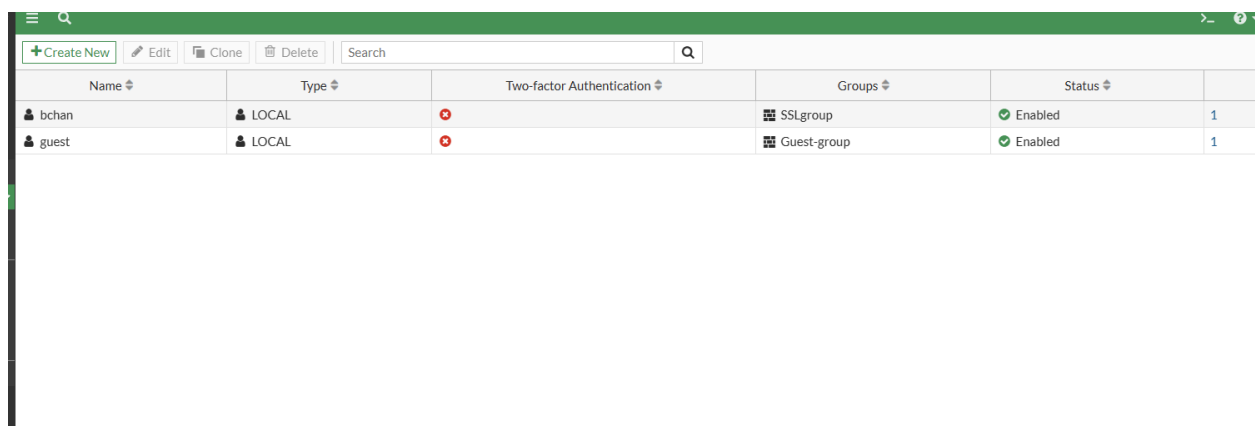
- Tunnel Template:** Site to Site - FortiGate
- Name:** FG1 to FG2
- Comments:** VPN: FG1 to FG2 (Created by VPN wizard)
- Network:** Remote Gateway: Static IP Address (10.0.11.2), Outgoing Interface: Lan2
- Authentication:** Authentication Method: Pre-shared Key
- Phase 2 Selectors:**

Local Address	Remote Address
FG1 to FG2	FG1 to FG2_remote

Additional Information on the right includes links to API Preview, References, IPsec VPNs, Guides, IPsec VPN Cookbook, VPN Setup on FortiClient, Configuring an IPsec, Documentation, Online Help, and Video Tutorials.

Configure Client-Based VPN for remote access

remote employee connectivity, a client-based Secure Sockets Layer (SSL) VPN was implemented on the primary firewall ,a Windows-based virtual machine was deployed inside PC1 with client user being assigned a BCIT address An external user account bchan was first created to handle remote user login authentication This user account was added to the correct user group policy SSLgroup An SSL-VPN portal was then created on listening port 8080, and configured to assign users a specific IP pool when connecting to the tunnel. The user group was added to the authentication mapping parameters during the SSL-VPN setup When configuring the stateful SSL policy on FG1, the source was assigned to the SSL group, allowing all incoming connections to route to internal targets Remote users are required to download the Fortinet VPN client software to connect to the internal network remote gateway setting to 192.168.1.99 over custom port 8080, and using the created user account and correct password, the user successfully connected to the internal firewall and verified network visibility by pinging internal users



The screenshot shows the Fortinet User Management interface. At the top, there is a green header bar with a search icon and a help icon. Below the header, there is a toolbar with buttons for '+ Create New', 'Edit', 'Clone', 'Delete', and a search bar. The main content area displays a table with the following columns: Name, Type, Two-factor Authentication, Groups, Status, and a final column with a count. The table contains two rows of data.

Name	Type	Two-factor Authentication	Groups	Status	
bchan	LOCAL	✖	SSLgroup	✔ Enabled	1
guest	LOCAL	✖	Guest-group	✔ Enabled	1

Group Name	Group Type	Members	Ref
uest-group	Firewall	guest	0
3lgroup	Firewall	bchan	2
SO_Guest_Users	Fortinet Single Sign-On (FSSO)		1

FortiGate - FG1 x | Untitled docum x | Overview | Splu x | Homepage - Br x | Splunk App - C x | apps for splunk x | Hamid Capston x | 192.168.1.11 x | Olive Cafe Burn x

← → ↻ Not secure https://192.168.1.99/ng/vpn/ssl/settings

Google Chrome isn't your default browser [Set as default](#)

FG1

- Dashboard
- Network
- Policy & Objects
- Security Profiles
- VPN
 - Overlay Controller VPN
 - IPsec Tunnels
 - IPsec Wizard
 - IPsec Tunnel Template
 - SSL-VPN Portals
 - SSL-VPN Settings**
 - SSL-VPN Clients
 - VPN Location Map
 - User & Authentication
 - WiFi & Switch Controller
 - System
 - Security Fabric
 - Log & Report

SSL-VPN Settings

Connection Settings

Enable SSL-VPN ☒

Listen on Interface(s) Internal

Listen on Port 8080

Server Certificate Fortinet_Factory

Tunnel Mode Client Settings

Address Range Automatically assign addresses Specify custom IP ranges

DNS Server Same as client system DNS Specify

Specify WINS Servers ☐

Authentication/Portal Mapping

[+ Create New](#) [Edit](#) [Delete](#) [Send SSL-VPN Configuration](#)

Users/Groups	Portal
SSLgroup	myportal
All Other Users/Groups	full-access

[Apply](#)

Additional Information

[API Preview](#)

[Edit in CLI](#)

SSL VPN Setup Guides

Web Mode

[Web Mode for Remote User](#)

Tunnel Mode

[Full Tunnel for Remote User](#)

[Split Tunnel for Remote User](#)

[Tunnel Mode Host Check](#)

Multi-Realm

[Multi-Realm](#)

Authentication

[Certificate Authentication](#)

[LDAP-Integrated Certificate Aut](#)

[FortiToken Mobile Push Authent](#)

[RADIUS on FortiAuthenticator](#)

[RADIUS and FortiToken Mobile F](#)

[Local User Password Policy](#)

[RADIUS Password Renewal on For](#)

[LDAP User Password Renew](#)

VPN Setup on FortiClient

[Configuring an SSL VPN Connect](#)

Troubleshooting

[Troubleshooting](#)

Documentation

[Online Help](#)

[Video Tutorials](#)

Security Rating Issues

[Valid HTTPS Certificate - SSL-V](#)

[Show Dismissed](#)

Edit Policy

Name ⓘ	SSL to DMZ		
Incoming Interface ⚠️	SSL-VPN tunnel interface (ssl.roo) ▼		
Outgoing Interface	internal ▼		
Source	all ✕	SSLgroup ✕	+ ✕
Destination	all ✕		
Schedule	always ▼		
Service	ALL ✕		
Action	☒ ACCEPT ☐ DENY		

Inspection Mode **Flow-based** Proxy-based

Firewall / Network Options

NAT ☒

IP Pool Configuration **Use Outgoing Interface Address** Use Dynamic IP Pool

Preserve Source Port ☐

Protocol Options **PROT** default ▼ ✎

Security Profiles

AntiVirus ☐

Web Filter ☐

DNS Filter ☐

Application Control ☐

File Filter ☐

SSL Inspection **SSL** no-inspection ▼ ✎

Logging Options

☐ Log Connections ☐ Log Sessions ☐ Log Data ☐ Log Alerts

Statistics (since last reset)

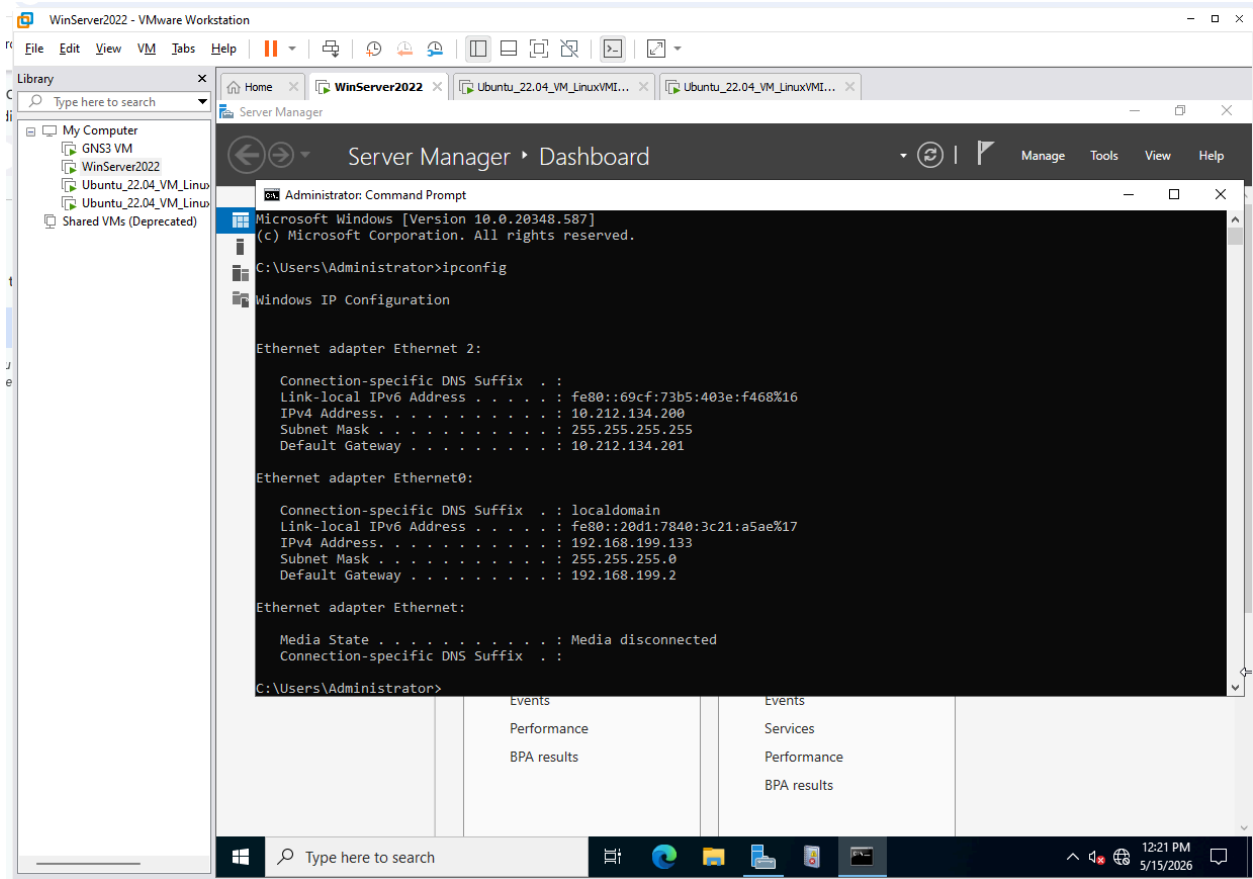
ID	4
Last used	N/A
First used	N/A
Active sessions	0
Hit count	0
Total bytes	0B
Current bandwidth	0 bps

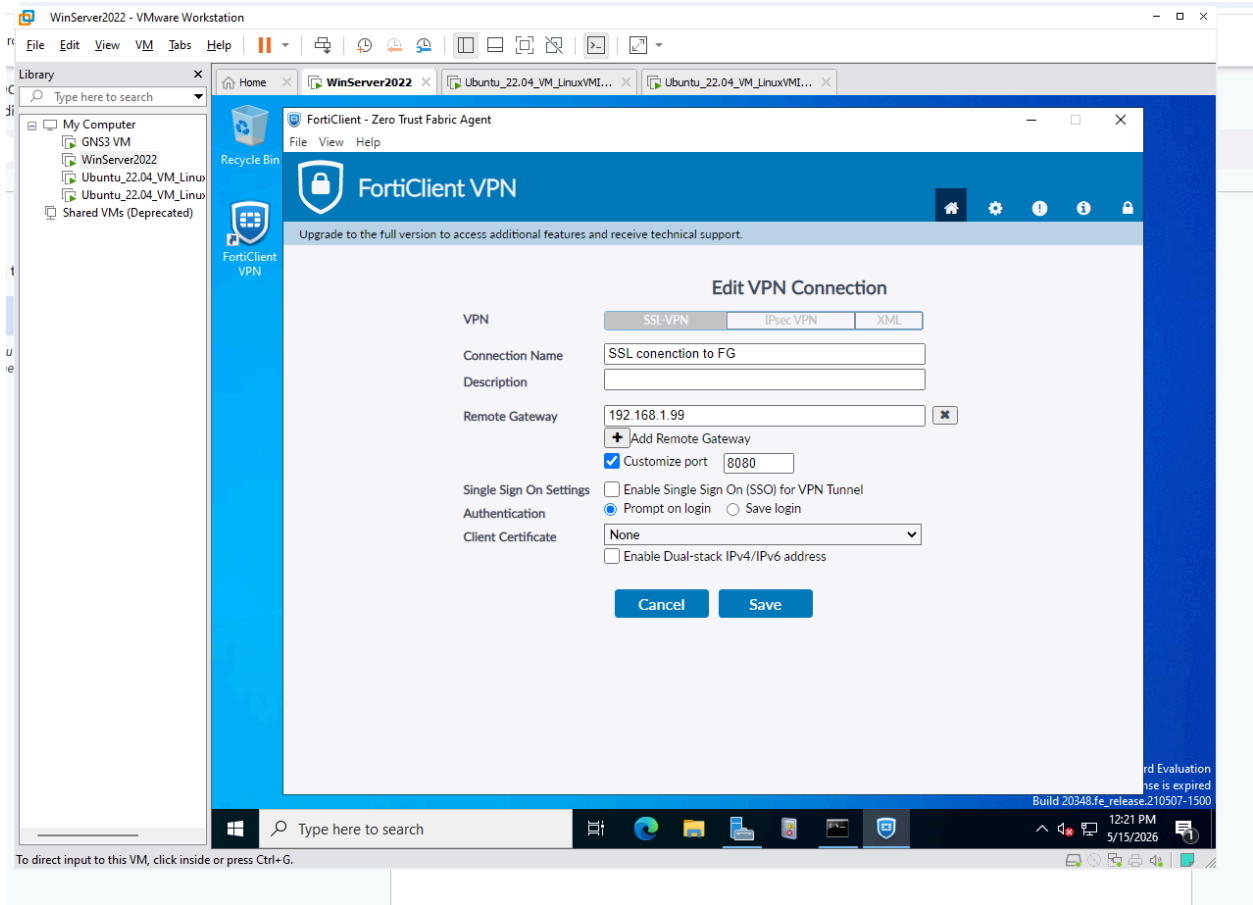
Clear Counters

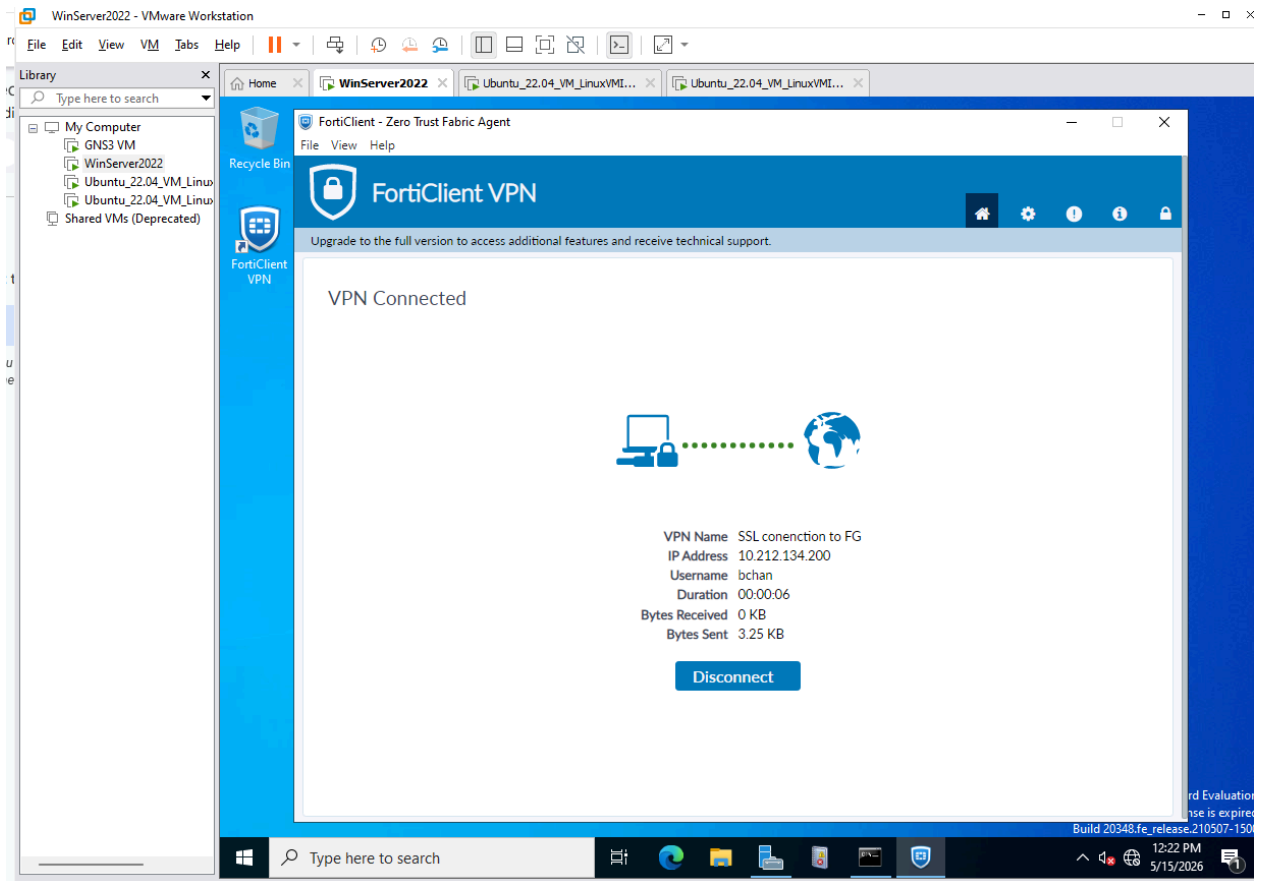
Additional Information

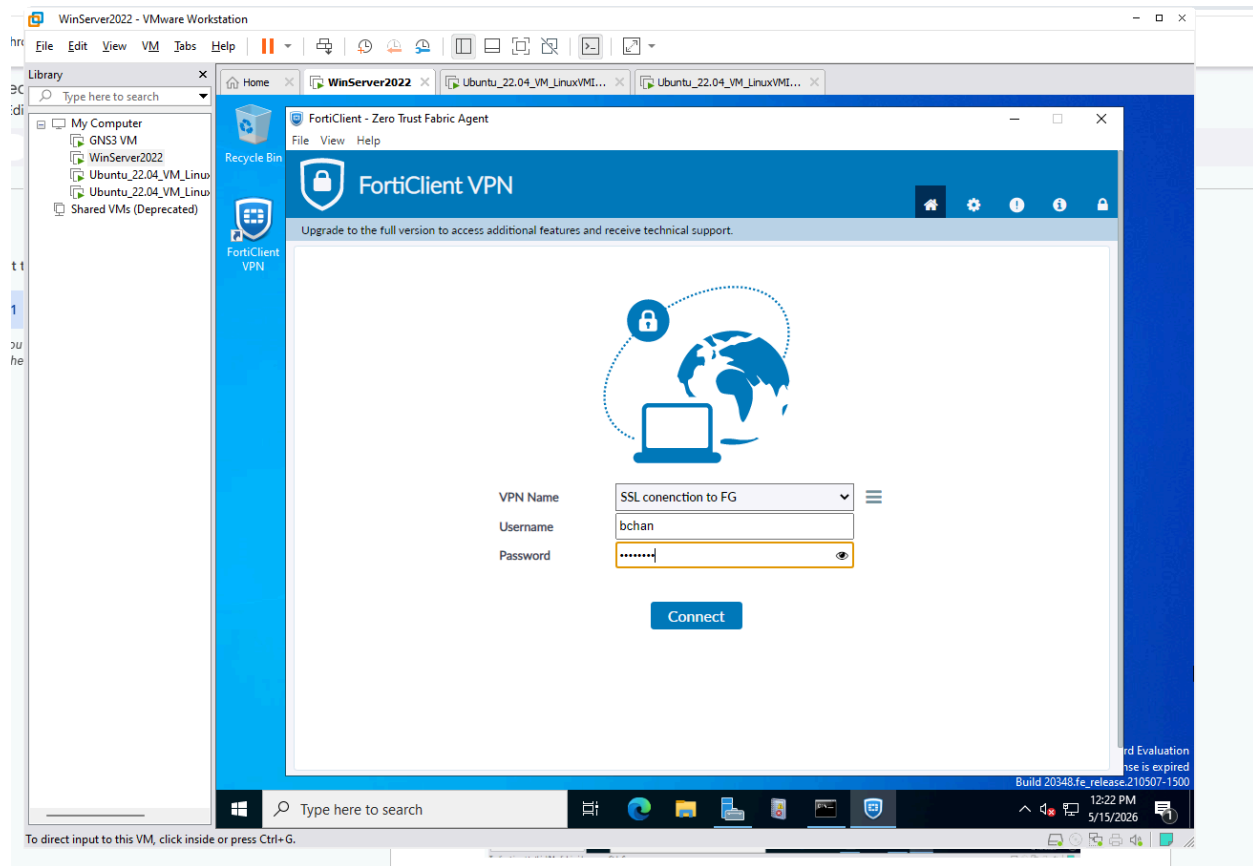
② Documentation

- [Online Help](#) ↗
- [Video Tutorials](#) ↗
- [Consolidated Policy Configuration](#)









Send all logs to Splunk

two attack SSH bruteforce attack and an Nmap port scan were simulated inside the environment using an Kali Linux Kali Linux virtual machine was built on PC1, and a secondary Kali Linux virtual machine was created inside PC2 to simulate an internal attack vector via SSH

After ensuring baseline network routing by verifying that the PC2 Kali Linux VM was able to successfully ping the PC1 Kali Linux VM, an brute force attack was launched. From the PC2 virtual machine, the login cracker Hydra was executed against the internal host target 192.168.1.10 hydra -l ubuntu -P pass.txt

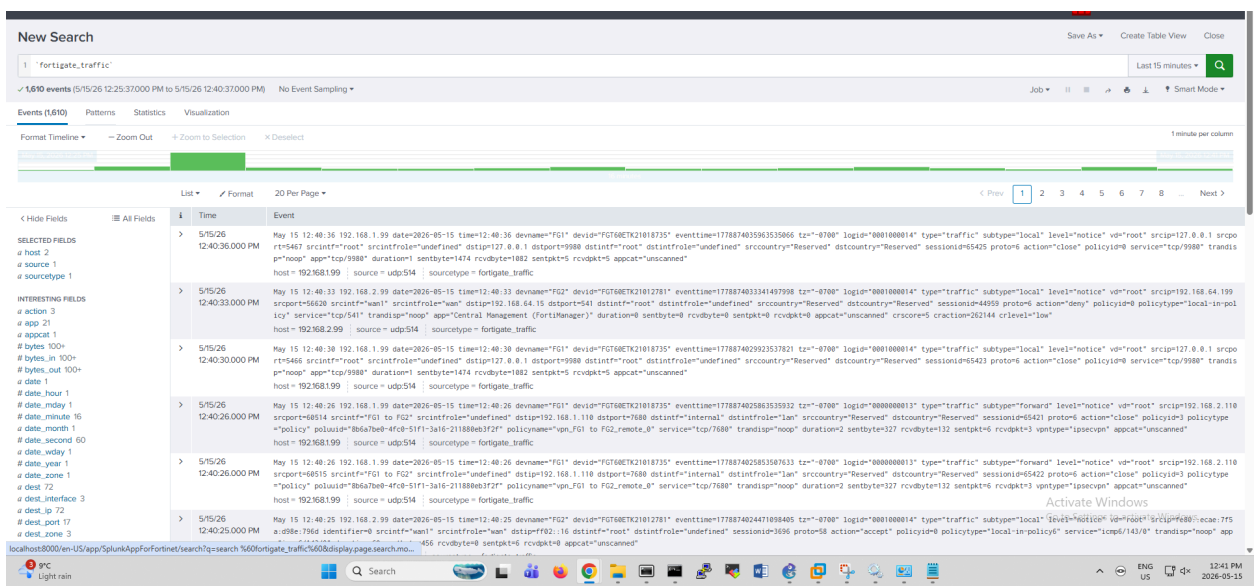
192.168.1.10 ssh -vV

shown in the documentation screenshots, Splunk successfully tracked the traffic lifecycle of the brute-force The logs provide evidence that the host at 192.168.2.10 successfully completed an unauthorized SSH authentication sequence into the target server at 192.168.1.10

The Kali Linux VM ciphered through the entries in the supplied password dictionary file pass.txt against the target host on PC1 It eventually succeeded in

finding the correct login credentials, allowing the attacker to authenticate and log in via SSH using the command `ssh ubuntu@192.168.1.10`

The Nmap network utility was run against the target system using service-version detection flags `nmap -sV 192.168.1.10`. The scan queried standard destination transport ports to check all available and open ports for access. The output demonstrated that an SSH service was openly available on the host, along with an HTTP service running on IP 192.168.1.11, which hosts the company's internal Apache web server. Event logs further showed that the attacker at 192.168.2.10 attempted to scan the network for active ports like SSH and HTTP. However, a pre-configured DoS Protection Policy set up on the firewall actively monitored these inbound scanning sweeps. The policy flagged the rapid connection spikes as malicious scanning anomalies and successfully blocked the incoming scans from executing fully.



DOS policy

Edit Policy

Name

DOS policy

Incoming Interface

FG1 to FG2

Outgoing Interface

Internal

Source

Attacker IP

Destination

Internal

Schedule

always

Service

ALL

Action

ACCEPT DENY

Log Violation Traffic

Comments

Write a comment...

0/1023

Enable this policy

Statistics (since last reset)

ID	7
Last used	1 second(s) ago
First used	11 minute(s) ago
Active sessions	0
Hit count	70
Total bytes	5.64 kB
Current bandwidth	0 bps

Clear Counters

Last 7 Days

Bytes

Additional Information

API Preview

Edit in CLI

Documentation

Online Help

Google Chrome isn't your default browser

Set as default

splunk-enterprise

Apps

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Messages

Settings

Activity

Help

Find

Search & Report

Logs for FG

Edit

Export

i	Time	Event
>	5/15/26 12:16:29.000 PM	May 15 12:16:29 192.168.1.99 date=2026-05-15 time=12:16:29 devname="FG1" devid="FGT68ETK21018735" eventtime=17788725894753153 tzone="0700" logid="0000000013" type="traffic" subtype="forward" level="notice" vdr="root" srcip=192.168.2.10 srqport=53652 srcintf="FG1 to FG2" srctintfrole="undefined" dstip=192.168.1.11 dstport=80 dstintf="Internal" dstintfrole="lan" srccountry="Reserved" dstcountry="Reserved" sessionid=63124 proto=6 action="deny" policyid=7 policytype="policy" poluuid="93b7ad38-5090-51f1-3a16-77b3a07c9ee9" policyname="DOS p licy" service="HTTP" transidsp="noop" duration=0 sentbyte=0 rcvbyte=0 sentpkt=0 rcvpkt=0 vporttype="ipsecvpn" appcat="unscanned" crscore=38 craction=131872 crlevel="high" host = 192.168.1.99 source = udp514 sourcetype = fortigate.traffic
>	5/15/26 12:16:29.000 PM	May 15 12:16:29 192.168.1.99 date=2026-05-15 time=12:16:29 devname="FG1" devid="FGT68ETK21018735" eventtime=17788725894742024 tzone="0700" logid="0000000013" type="traffic" subtype="forward" level="notice" vdr="root" srcip=192.168.2.10 srqport=59408 srcintf="FG1 to FG2" srctintfrole="undefined" dstip=192.168.1.11 dstport=443 dstintf="Internal" dstintfrole="lan" srccountry="Reserved" dstcountry="Reserved" sessionid=63111 proto=6 action="deny" policyid=7 policytype="policy" poluuid="93b7ad38-5090-51f1-3a16-77b3a07c9ee9" policyname="DOS p licy" service="HTTPS" transidsp="noop" duration=0 sentbyte=0 rcvbyte=0 sentpkt=0 rcvpkt=0 vporttype="ipsecvpn" appcat="unscanned" crscore=38 craction=131872 crlevel="high" host = 192.168.1.99 source = udp514 sourcetype = fortigate.traffic
>	5/15/26 12:16:27.000 PM	May 15 12:16:27 192.168.1.99 date=2026-05-15 time=12:16:27 devname="FG1" devid="FGT68ETK21018735" eventtime=177887258784515482 tzone="0700" logid="0000000013" type="traffic" subtype="forward" level="notice" vdr="root" srcip=192.168.2.10 srqport=59408 srcintf="FG1 to FG2" srctintfrole="undefined" dstip=192.168.1.11 dstport=443 dstintf="Internal" dstintfrole="lan" srccountry="Reserved" dstcountry="Reserved" sessionid=63111 proto=6 action="deny" policyid=7 policytype="policy" poluuid="93b7ad38-5090-51f1-3a16-77b3a07c9ee9" policyname="DOS p licy" service="HTTPS" transidsp="noop" duration=0 sentbyte=0 rcvbyte=0 sentpkt=0 rcvpkt=0 vporttype="ipsecvpn" appcat="unscanned" crscore=38 craction=131872 crlevel="high" host = 192.168.1.99 source = udp514 sourcetype = fortigate.traffic
>	5/15/26 12:16:27.000 PM	May 15 12:16:27 192.168.1.99 date=2026-05-15 time=12:16:27 devname="FG1" devid="FGT68ETK21018735" eventtime=177887258784515482 tzone="0700" logid="0000000013" type="traffic" subtype="forward" level="notice" vdr="root" srcip=192.168.2.10 srqport=53658 srcintf="FG1 to FG2" srctintfrole="undefined" dstip=192.168.1.11 dstport=80 dstintf="Internal" dstintfrole="lan" srccountry="Reserved" dstcountry="Reserved" sessionid=63118 proto=6 action="deny" policyid=7 policytype="policy" poluuid="93b7ad38-5090-51f1-3a16-77b3a07c9ee9" policyname="DOS p licy" service="HTTP" transidsp="noop" duration=0 sentbyte=0 rcvbyte=0 sentpkt=0 rcvpkt=0 vporttype="ipsecvpn" appcat="unscanned" crscore=38 craction=131872 crlevel="high" host = 192.168.1.99 source = udp514 sourcetype = fortigate.traffic
>	5/15/26 12:12:06.000 PM	May 15 12:12:06 192.168.1.99 date=2026-05-15 time=12:12:06 devname="FG1" devid="FGT68ETK21018735" eventtime=17788722586932293 tzone="0700" logid="0000000013" type="traffic" subtype="forward" level="notice" vdr="root" srcip=192.168.2.10 srqport=51146 srcintf="FG1 to FG2" srctintfrole="undefined" dstip=192.168.1.11 dstport=2845 dstintf="Internal" dstintfrole="lan" srccountry="Reserved" dstcountry="Reserved" sessionid=62559 proto=6 action="timeout" policyid=3 policytype="policy" poluuid="8b6a7beb-4fc8-51f1-3a16-211880eb3f2f" policyname=" p_FG1 to FG2_remote_8" service="SIP-MSMessenger" transidsp="noop" duration=0 sentbyte=0 rcvbyte=0 sentpkt=1 rcvpkt=0 vporttype="ipsecvpn" appcat="unscanned" host = 192.168.1.99 source = udp514 sourcetype = fortigate.traffic
>	5/15/26 12:12:06.000 PM	May 15 12:12:06 192.168.1.99 date=2026-05-15 time=12:12:06 devname="FG1" devid="FGT68ETK21018735" eventtime=1778872258693228128 tzone="0700" logid="0000000013" type="traffic" subtype="forward" level="notice" vdr="root" srcip=192.168.2.10 srqport=45962 srcintf="FG1 to FG2" srctintfrole="undefined" dstip=192.168.1.11 dstport=80 dstintf="Internal" dstintfrole="lan" srccountry="Reserved" dstcountry="Reserved" sessionid=62559 proto=6 action="timeout" policyid=3 policytype="policy" poluuid="8b6a7beb-4fc8-51f1-3a16-211880eb3f2f" policyname=" p_FG1 to FG2_remote_8" service="SIP-MSMessenger" transidsp="noop" duration=0 sentbyte=0 rcvbyte=0 sentpkt=1 rcvpkt=0 vporttype="ipsecvpn" appcat="unscanned" host = 192.168.1.99 source = udp514 sourcetype = fortigate.traffic
>	5/15/26 12:12:06.000 PM	May 15 12:12:06 192.168.1.99 date=2026-05-15 time=12:12:06 devname="FG1" devid="FGT68ETK21018735" eventtime=177887225869322793 tzone="0700" logid="0000000013" type="traffic" subtype="forward" level="notice" vdr="root" srcip=192.168.2.10 srqport=51354 srcintf="FG1 to FG2" srctintfrole="undefined" dstip=192.168.1.11 dstport=1131 dstintf="Internal" dstintfrole="lan" srccountry="Reserved" dstcountry="Reserved" sessionid=62554 proto=6 action="timeout" policyid=3 policytype="policy" poluuid="8b6a7beb-4fc8-51f1-3a16-211880eb3f2f" policyname=" p_FG1 to FG2_remote_8" service="tcp/1131" transidsp="noop" duration=0 sentbyte=0 rcvbyte=0 sentpkt=1 rcvpkt=0 vporttype="ipsecvpn" appcat="unscanned" host = 192.168.1.99 source = udp514 sourcetype = fortigate.traffic
>	5/15/26 12:12:06.000 PM	May 15 12:12:06 192.168.1.99 date=2026-05-15 time=12:12:06 devname="FG1" devid="FGT68ETK21018735" eventtime=177887225869321527 tzone="0700" logid="0000000013" type="traffic" subtype="forward" level="notice" vdr="root" srcip=192.168.2.10 srqport=59816 srcintf="FG1 to FG2" srctintfrole="undefined" dstip=192.168.1.11 dstport=4129 dstintf="Internal" dstintfrole="lan" srccountry="Reserved" dstcountry="Reserved" sessionid=62568 proto=6 action="timeout" policyid=3 policytype="policy" poluuid="8b6a7beb-4fc8-51f1-3a16-211880eb3f2f" policyname=" p_FG1 to FG2_remote_8" service="tcp/4129" transidsp="noop" duration=0 sentbyte=0 rcvbyte=0 sentpkt=1 rcvpkt=0 vporttype="ipsecvpn" appcat="unscanned" host = 192.168.1.99 source = udp514 sourcetype = fortigate.traffic

9°C

Search

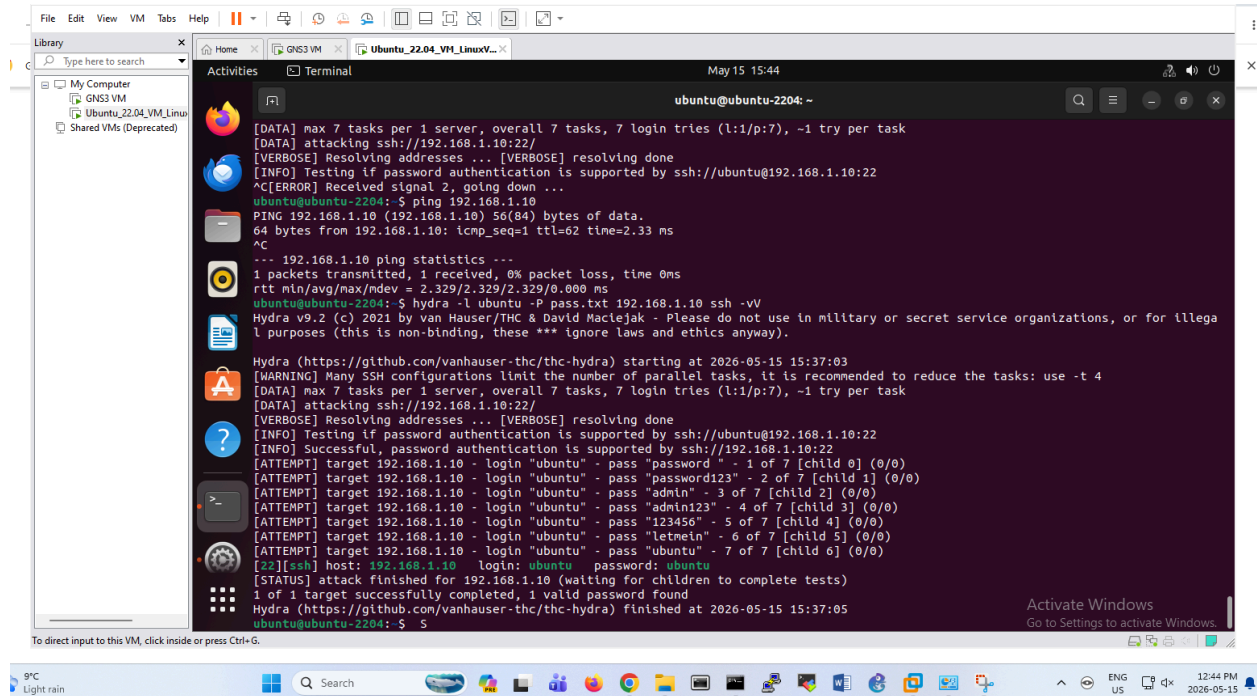
ENG

dx

12

ATTACK Simulation (SSH and NMAP)

PDF logs_for_fg1_skills-2026-05-15.pdf



APACHE server

